

1. What are Assignment Operators?

Assignment operators are used to assign values to variables. The basic assignment operator is `=`, and Python also provides shorthand operators like `+=`, `-=`, `*=`, `/=`, `%=`, `//=`, and `**=`. These operators make the code shorter and easier to read. They are commonly used to update the value of a variable.

2. What are Identity Operators?

Identity operators are used to check whether two variables refer to the same object in memory. Python provides two identity operators: **`is`** and **`is not`**. They compare the identity of objects instead of their values. These operators return either `True` or `False`.

3. What are Membership Operators?

Membership operators are used to check whether a value exists in a sequence such as a string, list, tuple, set, or dictionary. Python provides two membership operators: **`in`** and **`not in`**. They return `True` if the element is present; otherwise, they return `False`. They are commonly used in conditional statements and loops.

4. What are Bitwise Operators?

Bitwise operators perform operations on the binary (bit-level) representation of integers. Python provides operators such as `&`, `|`, `^`, `~`, `<<`, and `>>`. These operators are mainly used in low-level programming and optimization. They are less commonly used in everyday Python applications.

5. What is Operator Precedence?

Operator precedence is the order in which Python evaluates operators in an expression. Operators with higher precedence are evaluated before operators with lower precedence. Parentheses `()` can be used to change the order of evaluation. Understanding operator precedence helps avoid logical errors in programs.